

Austin Zhai

Calgary, AB | 587-890-8752 | austinhzhai@gmail.com | [Linkedin](#) | [GitHub](#) | [Website](#)

EDUCATION

University of British Columbia

Vancouver, BC

Bachelor of Applied Science in Computer Engineering

Graduation Date: April 2029

- Cumulative Average: 83.7% | GPA: 3.90/4.33 | Honors: Dean's List | Relevant Coursework: Intro to Computation in C: A-

EXPERIENCE

Operational Technology Engineering Intern

Calgary, AB

Galaxy Instrumentation

May 2026 – Present

- Audit CHEP OT infrastructure across 750+ global manufacturing sites at ~25 sites per week by executing pre and post migration network verification supporting the Perfect Plant initiative across FMS servers, PLCs, HMIs, and ADI devices
- Validate connectivity across ~7 Allen-Bradley PLCs per site by configuring drivers in RSLinx Classic Lite, verifying module and port configurations, and flagging communication failures for post-migration review
- Confirm ADI system health with heartbeat intervals under **300ms** by remotely connecting via Radmin, querying production history and alarm event tables in SSMS, and verifying data transfer speeds
- Cut manual audit setup time by ~15% by building a Python script integrated with Microsoft Power Automate to automate file and folder navigation across multi-site documentation workflows

IoT Cyber Defense Extern

Remote

Hydroficient

April 2026 – Present

- Eliminated unauthorized data injection across 3 simulated attack classes by designing and stress-testing a **5-layer** security pipeline using TLS, mTLS, HMAC signing, timestamp validation, and sequence counters across a 500-node IoT network
- Detect anomalous sensor readings in real time by implementing a Python/MQTT data acquisition pipeline and deploying an Isolation Forest model integrated into a Streamlit monitoring dashboard

Bilingual Data Analyst

Remote

TELUS Digital

March 2026 – Present

- Maintain ~90% accuracy across POI validity, search relevance, and location intent by evaluating **80+** bilingual map queries weekly in French and English to improve AI-powered mapping products used by millions globally

Operations Team Member

Vancouver, BC

UBC Sailbot

September 2025 – April 2026

- Grew social media engagement by ~80% and secured **18** active sponsors by independently managing a brand campaign across content planning, graphic design, and video production

Founder and Operator, Independent Haircutting Business

Calgary, AB

Self-employed

August 2024 – December 2025

- Generated **\$7,000+** revenue at ~85% retention by building a **50+** client haircutting business through Excel-based referral system

PROJECTS

Portfolio Vision

[Github](#)

React, JavaScript, Python, Supabase (PostgreSQL), Google Cloud, Vercel

May 2026

- Built and deployed a full-stack ETF portfolio decomposer reaching **400+** visitors in its first week by developing a recursive decomposition engine covering **500+** tickers with live USD/CAD conversion via Frankfurter API
- Eliminated **\$30+/month** in API costs and unlocked unlimited price fetching across TSX, NEO, and CSE by engineering a Vercel serverless proxy scraper to Yahoo Finance, bypassing paid API requirements entirely
- Secured multi-user portfolio saving by implementing Google OAuth and OTP email authentication with row-level security policies via Supabase Auth and Google Cloud OAuth 2.0

Servo Sonar Radar

[Github](#)

Arduino, C++, Processing

May 2026

- Built a real-time object detection system with a 180° sweep range and 3-mode state machine (auto-sweep, pause-on-detect, joystick control) by mounting an HC-SR04 ultrasonic sensor on a servo motor with millis()-based non-blocking timing
- Visualized live sonar data on a custom radar display rendering angle, distance, and proximity alerts in real time by implementing a Serial communication pipeline between Arduino and Processing at 9600 baud

SKILLS & INTERESTS

Languages: Python, C, C++, JavaScript, HTML/CSS

Technologies: React.js, Node.js, Vite, Tailwind CSS, Supabase (PostgreSQL), REST APIs, Git/GitHub, Power Automate

Interests: Personal Finance, Weight Training, Basketball/Sports, Piano